

# Host-parasite coevolution; origin and maintenance of sex

## Antagonistic coevolution

The costs of parasitism lead to **antagonistic coevolution** between hosts and parasites.

Hosts can evolve **resistance** (typically by closing the compatibility filter) or **tolerance** (focusing on resistance this week).

- **Red Queen** (cyclic)
  - new alleles arise frequently
  - *frequency-dependent selection*: rare alleles in the parasite, or host, have an advantage
  - resistance alleles tend to be *monomorphic*
  - different possible genetic systems: gene-for-gene, matching alleles, etc.

|               |   | Parasite fitness on host genotype $i$ |          |          |
|---------------|---|---------------------------------------|----------|----------|
|               |   | Parasite genotype                     |          |          |
| Host genotype | A | 1                                     | 2        | 3        |
|               | 1 | <i>B</i>                              | <i>O</i> | <i>O</i> |
|               | 2 | <i>O</i>                              | <i>B</i> | <i>O</i> |
|               | 3 | <i>O</i>                              | <i>O</i> | <i>B</i> |
|               |   | Inverse-matching alleles              |          |          |
|               |   | Parasite genotype                     |          |          |
| Host genotype | B | 1                                     | 2        | 3        |
|               | 1 | <i>O</i>                              | <i>B</i> | <i>B</i> |
|               | 2 | <i>B</i>                              | <i>O</i> | <i>B</i> |
|               | 3 | <i>B</i>                              | <i>B</i> | <i>O</i> |
|               |   | Gene-for-gene                         |          |          |
|               |   | Parasite genotype                     |          |          |
| Host genotype | C | 1                                     | 2        | 3        |
|               | 1 | <i>B</i>                              | <i>O</i> | <i>O</i> |
|               | 2 | <i>B</i>                              | <i>O</i> | <i>B</i> |
|               | 3 | <i>B</i>                              | <i>B</i> | <i>B</i> |

(Gibson and Lively (2019); see also Dybdahl and Storfer (2003))

- **matching alleles**: e.g. host has a **self-nonself recognition system** (typical in invertebrates); parasite succeeds if it matches (i.e. looks like host). Favors local adaptation, cycling/variation, dilution effects.
- **inverse matching alleles**: host matches many parasite signals (e.g. vertebrate antibody/antigen matching); parasite succeeds if it **does not** match any of the host antibodies. Doesn't favour rare host genotypes, anti-dilution effects.
- **gene-for-gene**: like inverse matching, but there is a "universal infecter" that doesn't match anything (i.e. infects everything) (Plants: R genes, Avr genes). Wikipedia:

Because there would be no evolutionary advantage to a pathogen keeping a protein that only serves to have it recognised by the plant, it is believed that the products of Avr genes play an important role in virulence [= ability to infect] in genetically susceptible hosts

Stahl et al. (1999): **Arms races** in a disease-resistance locus of *Arabidopsis*

- **trench warfare/arms race** (unidirectional)
  - resistance builds up until benefits balanced by costs
  - resistance alleles *polymorphic*
  - short-term *stabilizing* selection
  - long-term frequency cycling

## Antagonistic coevolution and sex

**Sex and variations**: we consider **dioecy** or **gonochory** (individuals are either male or female), but there are many variations: individuals may be **sequential** or **simultaneous** hermaphrodites. They may **self-fertilize** or **outcross** to different degrees.

**Costs of sex:** mating failure (vs. *reproductive assurance*), cost of males, **cost of meiosis** (when only half your genes make it into your offspring, you essentially pay a 50% fitness cost). Cost of outbreeding (breaking up co-adapted gene complexes: Goldberg et al. (2005)).

Simplest if we just think of cost of meiosis and advantages of recombination, although other costs and benefits do apply.

**Advantages of sex** (not necessarily parasite-related)

- **Muller’s ratchet** (fixation of deleterious alleles within lineages) [small (10–100 individuals) populations only];
- “Kondrashov’s hatchet” (Kondrashov 1993) (an analogue that works in large populations, with epistasis); natural selection is *more* effective at purging deleterious mutations. Also see Keightley and Otto (2006) on mutation purging.
- Hard (frequency-independent) habitat selection: allows sexual populations to inhabit transient niches that may not be available to asexuals; increases probability of making it through bad years
- Soft (frequency-dependent) or “tangled bank” habitat selection: allows sexual populations to avoid competition better, since they can use a wider variety of niches

**Requirements for RQ dynamics**

- heritable variation in host resistance to parasites
- heritable variation in parasite infectivity
- specificity

## Snails and trematodes

Lively, Dybdahl and others have studied the interaction of parasitism and sexual reproduction extensively in New Zealand lakes (they started collecting about 15 years ago) where there are mixed clonal (triploid) and sexual (diploid) populations of New Zealand mud snail, *Potamopyrgus antipodarum* which are parasitized by a castrating cestode, *Microphallus* spp. Genetic (electrophoretic) variability exists in hosts; gene flow of parasites is higher than gene flow of hosts, which helps the RQ work. (Asexual lineages are very common in snails.)

Primary theories for the variation in frequency of sexual snails among and within lakes:

- **Resistance tradeoffs:** genetic tradeoff between competitive ability and resistance to parasites
- **Reproductive assurance:** asexuals ensure reproduction and avoid costs of mating (assuming sexuals have some other advantage)
- **Lottery:** sexuals survive in a wider range of (micro)habitats
- **Tangled bank:** rare offspring of sexuals experience less competition
- **Red Queen:** sexuals (or uncommon clones) resist parasites better

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|                                       | findings                                                              |
|---------------------------------------|-----------------------------------------------------------------------|
| @livelyEvidence1987                   | more sexuals in lakes; sexuals correlated with parasites across lakes |
| @livelyAdaptation1989                 | parasites infect local hosts better, regardless of distance           |
| @livelyParthenogenesis1992            | no corr between pop density and sex freq                              |
| @jokelaSpatial1995a; @foxGenetic1996a | sex corr w/ parasites within lakes                                    |
| @dybdahlHost1997                      | time-lagged assoc betw parasites & common clones                      |
| @jokelaEvidence1997a                  | sex doesn’t outcompete asex in absence of parasites                   |
| @dybdahlHost1998                      | association between parasites and previously common clones            |
| @kristSpatial2000                     | snails in shallow water more susceptible                              |
| @livelyParasite2000                   | assoc between para & prev common local, but not non-local, hosts      |

## Potential problems for the Red Queen

- RQ may not work without strong parasite effects on host fitness
- can sexuals compete against a diverse set of clones?
- is a tiny bit of sex enough to maintain variation without losing the advantages of asexuality?
- why is there so much *obligate* sexuality/outcrossing?
- persistent asexual lineages (e.g. bdelloid rotifers, but see Schwander (2016))

## Other theories (Meirmans and Neiman 2006)

- **Muller’s ratchet plus RQ:** Parasites drive population fluctuations which tend to fix deleterious mutations in asexual lineages. *Predicts:* Frequent parasite-induced population crashes (removing parasites should remove the crashes); Relative fitness of the population should be *higher* after crashes
- **Tangled bank plus RQ:** *Mechanism:* parasite resistance determines competitive ability. *Predicts:* competitive outcomes (between common and rare clones, or between sexuals and asexuals) should vary in the presence and absence of parasites

In all of this, we need to be careful distinguishing the true effects of sexual reproduction. Ecologists tend to assume it produces “more variable” offspring, but this is not necessarily the case. What sex really does is to allow recombination of different genotypes ... what is the true relationship between sexual reproduction and variability? It depends on population size, how frequently asexual lineages are split off from the sexual population and how, etc. etc.. (Importance of **epistasis**: (Metzger et al. 2016))

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